TR5 Worked Examples: Exact Trig Values

Recording companion for the TR5 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Where is the point on the unit circle, and what are its coordinates? On the unit circle, $\cos\theta = x$ and $\sin\theta = y$, so $\tan\theta = y/x$, $\sec\theta = 1/x$, $\csc\theta = 1/y$ and $\cot\theta = x/y$. For a special angle, the reference angle gives the size of the coordinates and the quadrant gives their signs. From one coordinate, $x^2 + y^2 = 1$ gives the other, with its sign from the quadrant. A zero in a denominator means the value is undefined.

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Example 1: Sine of 150 Degrees

$$\sin(150°)$$

Question

What is the exact value?

- **A.** $-1/2$
- **B.** $(\sqrt{3})/2$
- **C.** $1/2$
- **D.** $-(\sqrt{3})/2$

Answer and reasoning

150° is in Quadrant II, 30° short of 180°, so its reference angle is 30°. $\sin 30° = 1/2$, and sine (the *y*-coordinate) is positive in Quadrant II: $\sin 150° = 1/2$. $-1/2$ gives the wrong sign; $(\sqrt{3})/2$ is $\sin 60°$, from using the wrong reference angle or the cosine of 30°; and $-(\sqrt{3})/2$ makes both slips. ***1/2*.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 2: Cosine of 5π/4

$$\cos(5\pi/4)$$

Question

What is the exact value?

- **A.** $-(\sqrt{2})/2$
- **B.** $(\sqrt{2})/2$
- **C.** $-\sqrt{2}$
- **D.** $\sqrt{2}$

Answer and reasoning

$5\pi/4=225°$, in Quadrant III, with reference angle $\pi/4$. $\cos\pi/4=(\sqrt{2})/2$, and cosine (the x-coordinate) is negative in Quadrant III: $-(\sqrt{2})/2$. $(\sqrt{2})/2$ misses the sign; $-\sqrt{2}$ is $1/x$, the secant; and $\sqrt{2}$ makes both slips. **$-(\sqrt{2})/2$.**

Example 3: Tangent of 2π/3

$$\tan(2\pi/3)$$

Question

What is the exact value?

- **A.** $\sqrt{3}$
- **B.** $-\sqrt{3}$
- **C.** $-(\sqrt{3})/3$
- **D.** $(\sqrt{3})/3$

Answer and reasoning

$2\pi/3 = 120°$, in Quadrant II, with reference angle 60°. The point is $(-1/2, (\sqrt{3})/2)$, so $\tan = y/x = (\sqrt{3}/2)/(-1/2) = -\sqrt{3}$. $\sqrt{3}$ misses that x is negative; $-(\sqrt{3})/3$ is x/y, the cotangent, or $\tan 30°$ with a minus; and $(\sqrt{3})/3$ makes both slips. **$-\sqrt{3}$.**

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Example 4: Secant of 270 Degrees

$$\sec(270°)$$

Question

What is the exact value?

A. Undefined
B. Zero
C. One
D. Negative one

Answer and reasoning

At 270° the point on the unit circle is $(0,-1)$, straight down. $\sec = 1/x = 1/0$, and dividing by 0 is undefined. CheckIt's key prints undefined. Zero is $\cos 270°$, the x-coordinate itself; one treats $1/0$ as 1; and negative one is $\sin 270°$, or $\csc 270°$. **Undefined.**

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Example 5: Tangent from a Point

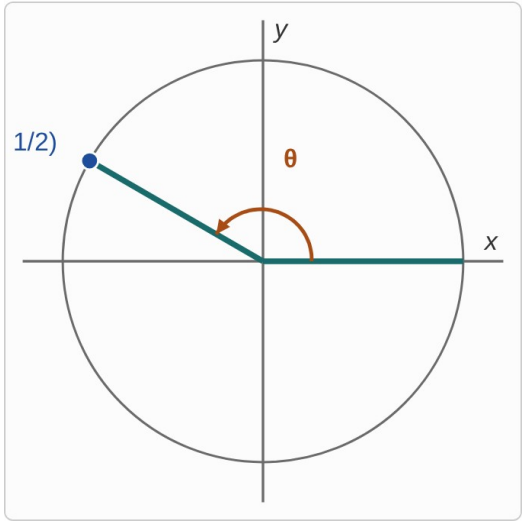


Figure: A unit circle with an angle θ in standard position. The terminal side points up and to the left, into Quadrant II, and meets the circle at the point labeled (−√3/2, 1/2).

$$(-(\sqrt{3})/2, 1/2)$$

Question

The terminal side of θ meets the unit circle at this point. What is tan θ?

- **A.** $-\sqrt{3}$
- **B.** $-(\sqrt{3})/3$
- **C.** $(\sqrt{3})/3$
- **D.** $\sqrt{3}$

Answer and reasoning

$\tan\theta = y/x = (1/2)/(-\sqrt{3}/2) = -1/\sqrt{3} = -(\sqrt{3})/3$. The halves cancel, and multiplying top and bottom by $\sqrt{3}$ clears the root from the bottom. $-\sqrt{3}$ is x/y, the cotangent; $(\sqrt{3})/3$ drops the sign of x; and $\sqrt{3}$ makes both slips. ***−(√3)/3*.**

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Example 6: Cosecant from a Point

$$((\sqrt{2})/2, -(\sqrt{2})/2)$$

Question

The terminal side of θ meets the unit circle at this point. What is csc θ?

- **A.** $\sqrt{2}$
- **B.** $-(\sqrt{2})/2$
- **C.** $(\sqrt{2})/2$
- **D.** $-\sqrt{2}$

Answer and reasoning

$\csc\theta = 1/y = 1/(-\sqrt{2}/2) = -2/\sqrt{2} = -\sqrt{2}$. $\sqrt{2}$ drops the sign of y; $-(\sqrt{2})/2$ is y itself, the sine, without taking the reciprocal; and $(\sqrt{2})/2$ is x, the cosine. **$-\sqrt{2}$.**

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Example 7: Secant from a Point

$$(-1/2, -(\sqrt{3})/2)$$

Question

The terminal side of θ meets the unit circle at this point. What is sec θ?

- **A.** -2
- **B.** 2
- **C.** $-(2\sqrt{3})/3$
- **D.** $(2\sqrt{3})/3$

Answer and reasoning

$\sec\theta = 1/x = 1/(-1/2) = -2$. 2 drops the sign; $-(2\sqrt{3})/3$ is $1/y$, the cosecant; and $(2\sqrt{3})/3$ is the cosecant with its sign dropped too. ***−2*.**

Example 8: On the Axis

$$(0,-1)$$

Question

The terminal side of θ meets the unit circle at this point. What is cot θ?

A. −1
B. 0
C. Undefined
D. 1

Answer and reasoning

cot$\theta = x/y = 0/(-1) = 0$. Zero on top is fine; only zero on the bottom is undefined. Undefined is tan$\theta = y/x = (-1)/0$, the reciprocal; -1 is y, the sine; and 1 is $(-1)/(-1)$, as if x were -1 too. **0.**

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Example 9: Quadrant II

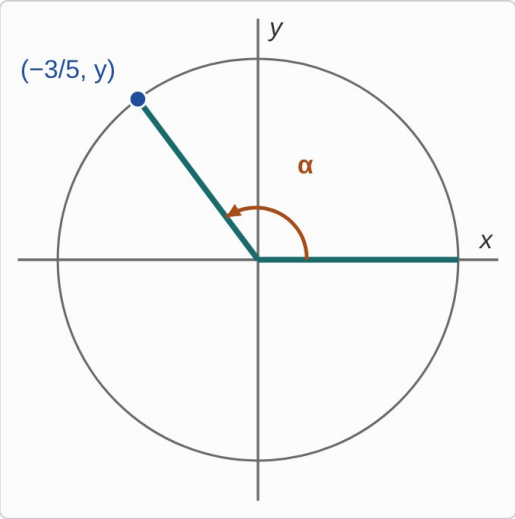


Figure: A unit circle with an angle α in standard position. The terminal side points up and to the left, into Quadrant II, and meets the circle at the point labeled (−3/5, y).

Question

The point (x, y) on the unit circle is in Quadrant II with x = −3/5, and α is its angle. What is tan α?

- **A.** $-4/3$
- **B.** $4/3$
- **C.** $-3/4$
- **D.** $3/4$

Answer and reasoning

$y^2=1-9/25=16/25$, so $y=\pm 4/5$, and in Quadrant II y is positive: $y=4/5$. Then $\tan\alpha=y/x=(4/5)/(-3/5)=-4/3$. It is the 3, 4, 5 triple. $4/3$ drops the sign of x (or makes y negative too); $-3/4$ is x/y, the cotangent; and $3/4$ makes both slips. ***−4/3*.**

Example 10: Quadrant IV

$$x=5/13$$

Question

The point (x, y) on the unit circle is in Quadrant IV with this x-value, and α is its angle. What is tan α?

- **A.** $12/5$
- **B.** $-5/12$
- **C.** $-12/5$
- **D.** $5/12$

Answer and reasoning

$y^2=1-25/169=144/169$, so $y=\pm 12/13$, and in Quadrant IV y is negative: $y=-12/13$ (the 5, 12, 13 triple). Then $\tan\alpha=y/x=(-12/13)/(5/13)=-12/5$. Tangent is negative in Quadrant IV. CheckIt's answer key shows $12/5$ for problems like this, because its generator puts the point in Quadrant I; trust the quadrant. $12/5$ makes y positive; $-5/12$ is x/y, the cotangent; and $5/12$ makes both slips. ***−12/5*.**

Example 11: Quadrant III

$$x=-8/17$$

Question

The point (x, y) on the unit circle is in Quadrant III with this x-value, and α is its angle. What is cot α?

- **A.** $-8/15$
- **B.** $15/8$
- **C.** $-15/8$
- **D.** $8/15$

Answer and reasoning

$y^2=1-64/289=225/289$, so $y=-15/17$, negative in Quadrant III (the 8, 15, 17 triple). Then $\cot\alpha=x/y=(-8/17)/(-15/17)=8/15$: two negatives make a positive, as cotangent always is in Quadrant III. $-8/15$ makes y positive; $15/8$ is y/x, the tangent; and $-15/8$ makes both slips. ***8/15*.**

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Example 12: Quadrant I

$$y=24/25$$

Question

The point (x, y) on the unit circle is in Quadrant I with this y-value, and α is its angle. What is sec α?

- **A.** *25/7*
- **B.** *7/25*
- **C.** *24/7*
- **D.** *7/24*

Answer and reasoning

$x^2=1-576/625=49/625$, so $x=7/25$, positive in Quadrant I (the 7, 24, 25 triple). Then $\sec\alpha=1/x=25/7$. *7/25* is *x* itself, the cosine; *24/7* is *y/x*, the tangent; and *7/24* is *x/y*, the cotangent. ***25/7*.**

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Example 13: Slip 1

$$\cos(210°)=\cos(30°)=(\sqrt{3})/2$$

Question

A student found this exact value. What is the mistake, if any?

A. There is no mistake.
B. In Quadrant III, x is negative: $-\sqrt{3}/2$.
C. The reference angle is 60°, so the value is 1/2.
D. Cosine of 210° is undefined, since 210° > 180°.

Answer and reasoning

The reference angle is right: $210°-180°=30°$, so the size is $(\sqrt{3})/2$. But 210° is in Quadrant III, where the x-coordinate is negative, so $\cos 210°=-(\sqrt{3})/2$. The reference angle is measured to the x-axis, so it is 30°, not 60°; and cosine is defined for every angle. ***−(√3)/2*.**

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Example 14: Slip 2

$$\tan(\pi/2)=\sin(\pi/2)/\cos(\pi/2)=1/0=0$$

Question

A student found this exact value. What is the mistake, if any?

A. There is no mistake.
B. Sine and cosine are swapped, so it is 0/1 = 0.
C. Dividing by 0 is not 0: tan(π/2) is undefined.
D. 1/0 is 1, so tan(π/2) = 1.

Answer and reasoning

At $\pi/2$ the point is $(0,1)$, so $\tan=y/x=1/0$, and division by 0 has no value: $\tan\pi/2$ is undefined. CheckIt's key prints undefined for this. The student had sine and cosine the right way: $\sin\pi/2=1$ and $\cos\pi/2=0$, so swapping them gives cotangent, not tangent; and $1/0$ is not 1 or anything else. **Undefined.**

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Example 15: Slip 3

$$\csc(7\pi/4)=1/(-\sqrt{2}/2)=-\sqrt{2}$$

Question

A student found this exact value. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** Cosecant uses x: the value is √2.
- **C.** Its reference angle is π/4, so the value is √2.
- **D.** The answer must be rationalized: −√2/2.

Answer and reasoning

$7\pi/4=315°$, in Quadrant IV, with point $((\sqrt{2})/2,-(\sqrt{2})/2)$. $\csc=1/y=-2/\sqrt{2}=-\sqrt{2}$. The student is right. Cosecant is the reciprocal of y, and $1/x$ is secant; the reference angle gives only the size, and the sign comes from Quadrant IV, where y is negative; and $-\sqrt{2}$ has no root in a denominator, while $-(\sqrt{2})/2$ is the sine, a different value. **There is no mistake.**

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Example 16: Slip 4

$$(-1/2,(\sqrt{3})/2): \quad \sec\theta = 2/\sqrt{3} = (2\sqrt{3})/3$$

Question

A student found sec θ for the angle whose unit-circle point is shown. What is the mistake, if any?

A. There is no mistake.
B. It should not be rationalized; leave it as $2/\sqrt{3}$.
C. Secant is y/x: the value is $-\sqrt{3}$.
D. Secant is 1/x, so the value is -2.

Answer and reasoning

$\sec\theta = 1/(\cos\theta) = 1/x = 1/(-1/2) = -2$. The student used $1/y$, which is the cosecant. Rationalizing is fine and equal, but it is the wrong function either way; and $y/x = -\sqrt{3}$ is the tangent. **-2.**

Example 17: Slip 5

$$y=12/13,\ \text{Quadrant II}$$
$$x=\sqrt{(1-144/169)}=5/13$$
$$\cos\alpha=5/13$$

Question

A student found cos α for a unit-circle point in Quadrant II. What is the mistake, if any?

A. There is no mistake.
B. In Quadrant II, x is negative: −5/13.
C. cos α is the y-value, so it is 12/13.
D. x = 1 − 12/13 = 1/13, so cos α = 1/13.

Answer and reasoning

The size of x is right: $\sqrt{(25/169)}=5/13$. But the square root gives only the size, and in Quadrant II x is negative, so $x=-5/13$ and $\cos\alpha=-5/13$. Cosine is the x-value, and $12/13$ is the sine; and $x^2+y^2=1$ works with squares, so $1-12/13$ is not x. ***−5/13*.**